

Autoimmunity

- Breaking of self-tolerance
- Involve

* tolerance: state of unresponsiveness (no immune response)
* tolerogene: antigen that cause tolerance
* self tolerance: negative selection

- Autoantibodies that activate complement system and ADCC mechanism

cell mediated immunity

- CMI that activate CD4+ and CD8+ cells

* breaking / disruption of self-tolerance causes autoimmune diseases

* autoimmune diseases:

- 1- antibody becomes autoantibody and react with self antigens
- 2- T cells becomes auto T cells

Demonstration of Tolerance

Medewar's Neonatal tolerance: exp.

TOLERANCE

A state of Unresponsiveness to
self Ag

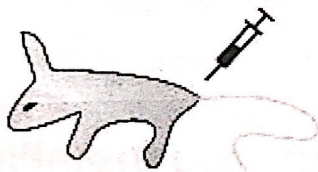
Tolerogen: any substance that
cause tolerance

Demonstration of Tolerance

*negative selection
starts in a fetus

Medewar's Neonatal tolerance expt

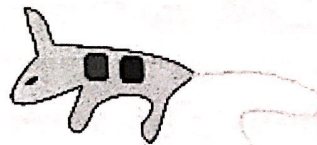
Week 0



صبيك الولادة
inject neonatal mouse (strain A)
with strain B bone marrow

حقن BM ل strain ج غريبة

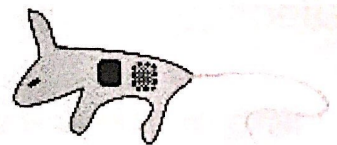
Week 6



graft skin from strain B
and strain C

زراعة skin من ال strain
المزوجة و strain افترى غريبة

Week 7



Strain B graft is accepted
Strain C graft is rejected

The mouse has **ACQUIRED** specific tolerance to strain B

*AMP is similar universally

Oral Tolerance (explains why we don't develop immune response against food)

- The gastrointestinal tract ^{الجهاز الهضمي} is the largest immunologic organ in the body. It is constantly bombarded by dietary proteins.
- Despite the extent of protein exposure, very few patients have food allergies because of development of oral tolerance to these antigens. Antigens might be taken up by overlying Peyer's patches, dendritic cells, or epithelial cells.
- Different cells of the immune system participate in oral tolerance induction, with **regulatory T cells** being the most important. *Treg excrete inhibitory cytokines CD10, TGFB
- Several factors can influence tolerance induction: Some are antigen related, and others are inherent to the host. Disturbances of oral tolerance have been described in food hypersensitivity.

Immune privilege Sites

- Do not develop i.r. to pathogens or tumors
- include eye, testes, ovary, brain and placenta
- corneal transplant: no need for tissue matching and immunosuppressive therapy

why is there no immune response in immune privilege sites?

a. lack of lymphatic drainage ** the immune response in these sites is done by filtration (T cells do not reach these sites)*

b. brain barrier

c. release of cytokines IL-10 and TGF

d. Expression of FasL: ** privilege sites has FasL * Fas - FasL → death signal*

human corneal endothelial cells and retinal epithelial cells have FasL

procedure: transplantation between two individuals of the same species (human → human, mouse → mouse)

- Mouse: allogeneic transplant → No rejection

gld/gld mice (lack FasL) → Allograft rejection

gold

homozygous, recessive

** FasL is responsible for acceptance or rejection*

Causes of Autoimmunity

(causes of breakage of self-tolerance)

①

- **Genetic factors:**

Polygenic: No individual gene is sufficient to elicit the disease.

Heterogenous system has different genes

Genetic predisposition: Associated with HLA: MS w DR2, MG w DR3, RA w DR4

there are several diseases associated with HLA:

myasthenia gravis (MG) with DR3

~~RA~~ rheumatoid arthritis (RA) with DR4

multiple sclerosis (MS) with DR2

②

Environmental factors

a. **Release of sequestered Ags** ^{ال Ag المحبأ} due to an accident or infection ^{when a privilege site comes out of place due to accident and blood mixes with Ags it strats to be recognized as foreign and an immune response is developed against the privilege site}
chondrocytes Ag of cartilage ^{ال Ag الكوندروسيت}

Cardiac Ags as myocarditis, ^{→ when the heart is recognized as a foreign Ag it's called myocarditis}

b. **Molecular mimicry** ^(the main cause of AID)

^{*mimicry between bacterial/viral Ag with self Ag}

Abs or T cells specific to microbial Ags cross-react with self Ag. (Epitope homology)

^{* when mimicry Ags enter the body, immune system will recognize them as foreign and attack them plus the self Ags that's been mimiced}

^{روماتيزم القلب} e.g. ^{streptococcus} Rheumatic fever follow *S. pyogenes* infection with protein M that have AutoAbs to cardiac myosin Ag.

[↓] **IDDM** ^{in *S. pyogenes*} **GAD** ^{enzyme in pancreas} w **Coxsackievirus** ^{causes diabetes in heart} [↓] ^{symptomless, not dangerous}

monoclonal activation: each epitope becomes a clone
polyclonal activation: activate everything (including energized cells)

c. **Polyclonal**: Nonspecific trigger many B and T cell clones including autoreactive cells.

e.g. SuperAg, LPS (Gm -), bac lipoteichoic acid (Gm +), hypomethylated CpG Bac DNA

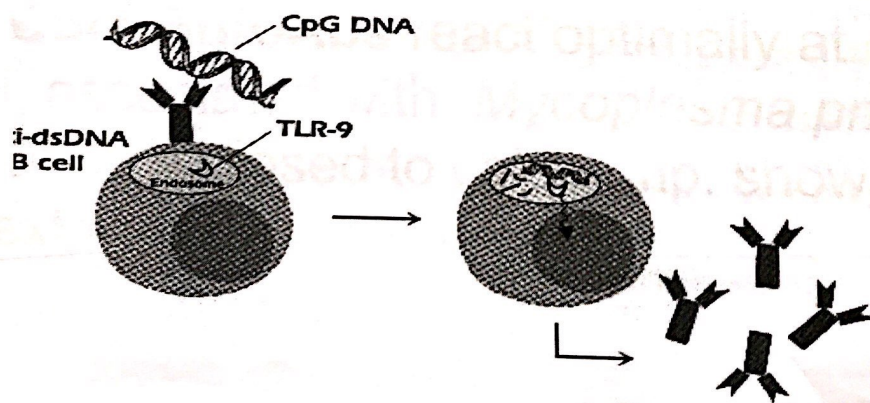
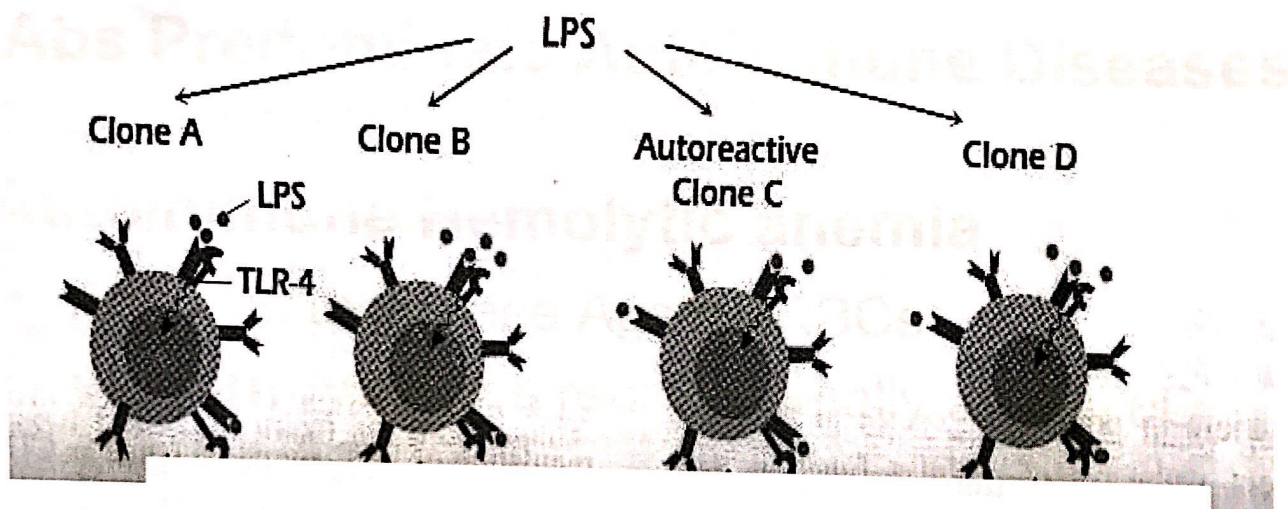
d. **Drugs and Hormones**

SLE is more common in females due to estrogens

Penicillin bind protein on RBC (lysis of RBCs)

α -methyl Dopa (antihypertensive drug)

Self limiting



Abs Predominate Autoimmune Diseases

① Autoimmune Hemolytic anemia

- AutoAbs to surface Ags on RBCs (self)

a. **Warm AutoAbs** react optimally at $> 37^{\circ}\text{C}$, IgG, Rh Ag *Rh on RBCs is recognized as foreign Ag, thus RBCs are lysed

b. **Cold AutoAbs** react optimally at $< 37^{\circ}\text{C}$, IgM, Agl, associated with *Mycoplasma pneumonia*. Patients exposed to cold temp. show symptoms in extremities الاصابع

*Agl on RBCs is recognized as foreign Ag, thus RBCs are lysed

② Myasthenia gravis

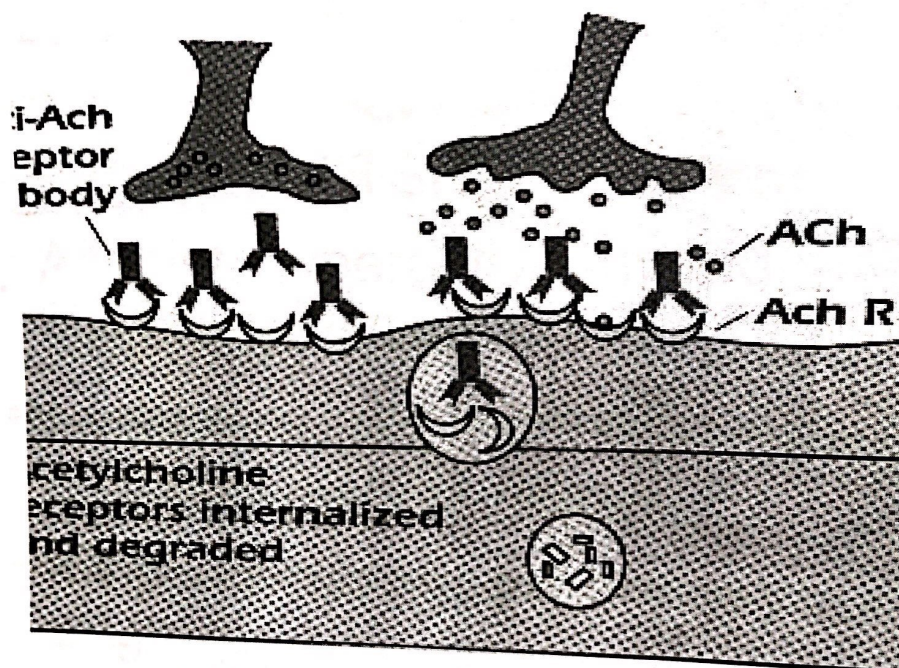
(muscles) (ضعف)

* normal individuals don't have Abs against ACh, but (auto Abs)

when there are Abs against the receptor, most of the ACh will be inhibited & Ab will bind the receptor and there's no place left for ACh thus no communication between nervous system and the muscle)

- Abs against Acetylcholine Receptor
- Act as antagonist that blocks binding the R
- Muscle weakness, inhibition of nerve impulse, problem in chewing and swallow and respiration
- Linked w thymoma (hypertrophy of thymus): removal of thymus lead to regression of the diseases. (cross-reaction thymus and Ag R). ^{thymus ال غدة}
- Animals---injection with AChR---→ induce MG ← ^{animal models (ex. rat)} ^{من}
AChR from Torpedo fish

Graves Disease



Myasthenia Gravis

B.

③ Graves Disease

- ^{antibodies} Abs against ^{hormone} TSH R ^{Thyroid stimulating receptor}
- Agonist: hyperactive thyroid ^{when there's more than enough thyroid} (Hyperthyroidism) ^{* thyroid gets its demands from pituitary gland in the brain that release TSH hormone}
^{* TSH stimulate thyroid}
^{* TSH + TSH R → Thyroid}
^{* thyroid → T₃, T₄}
^{* 3, 4 tyrosine molecules with tyrosine}
^{* thyroid regulate metabolism of carbohydrates, proteins, etc.}
- ^{symptoms:} Nervousness, Wt loss, increase in metabolism, bulging eyes ^{بالعينين فيها عينان كبريت}
- Pregnant mother give to the fetus through Abs ^(inherited disease)

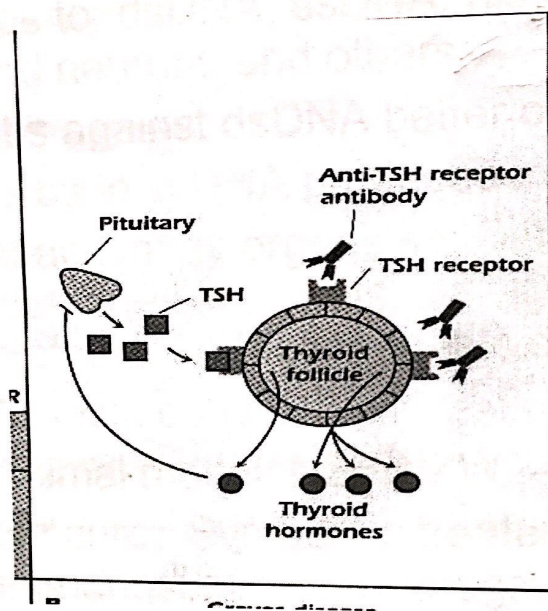


Fig. 7.12 The typical clinical appearance of Graves' disease includes exophthalmos and lid retraction, as seen in the

Systemic Lupus Erythematosus SLE

- Abs to dsDNA, ssDNA, histones, ribonucleoprotein, RBC and neurons and others (lymphocytes, T cells)
- Abs against dsDNA better correlates w the disease.
- Abs to ssDNA produced in normal individual
- Attack many organs so IC that trapped in glomerulus (damage skin, joints, kidney, heart), hematuria, proteinuria. Most patients die from kidney damage
- Associated w EBV (epstein virus) * a virus that exist in 95% of people
- Animal model: NZB/NZW (newzealand black/white mice)
- No cure. Symptoms treated with anti-inflammatory and immunosuppressive drugs.

↓
تخفيف الـ IIS



T cells predominate

الخلايا تي هي الغالبة

Hashimoto's thyroiditis

* thyroiditis → damage of thyroid

- Destruction of thyroid cells-Hypothyroidism-----goiter

تضخم الغدة الدرقية

- AutoAbs to thyroglobulin

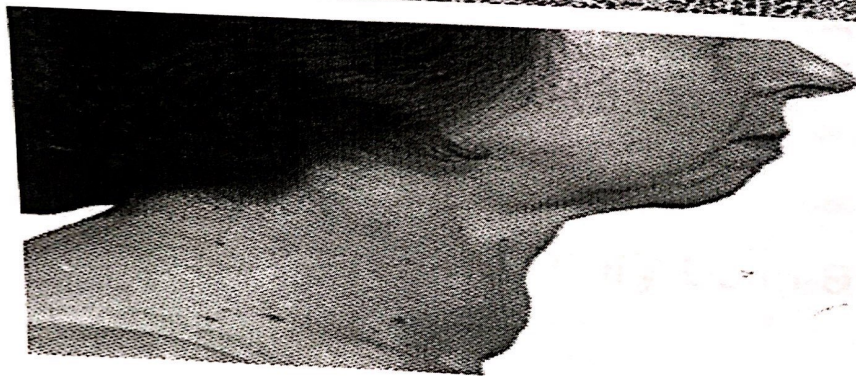
* tyrosine comes from a huge protein that breaks to make tyrosine, that protein is thyroglobulin in thyroid globulin

- Cold intolerance, wt gain

* Abs against thyroglobulin

Multiple sclerosis

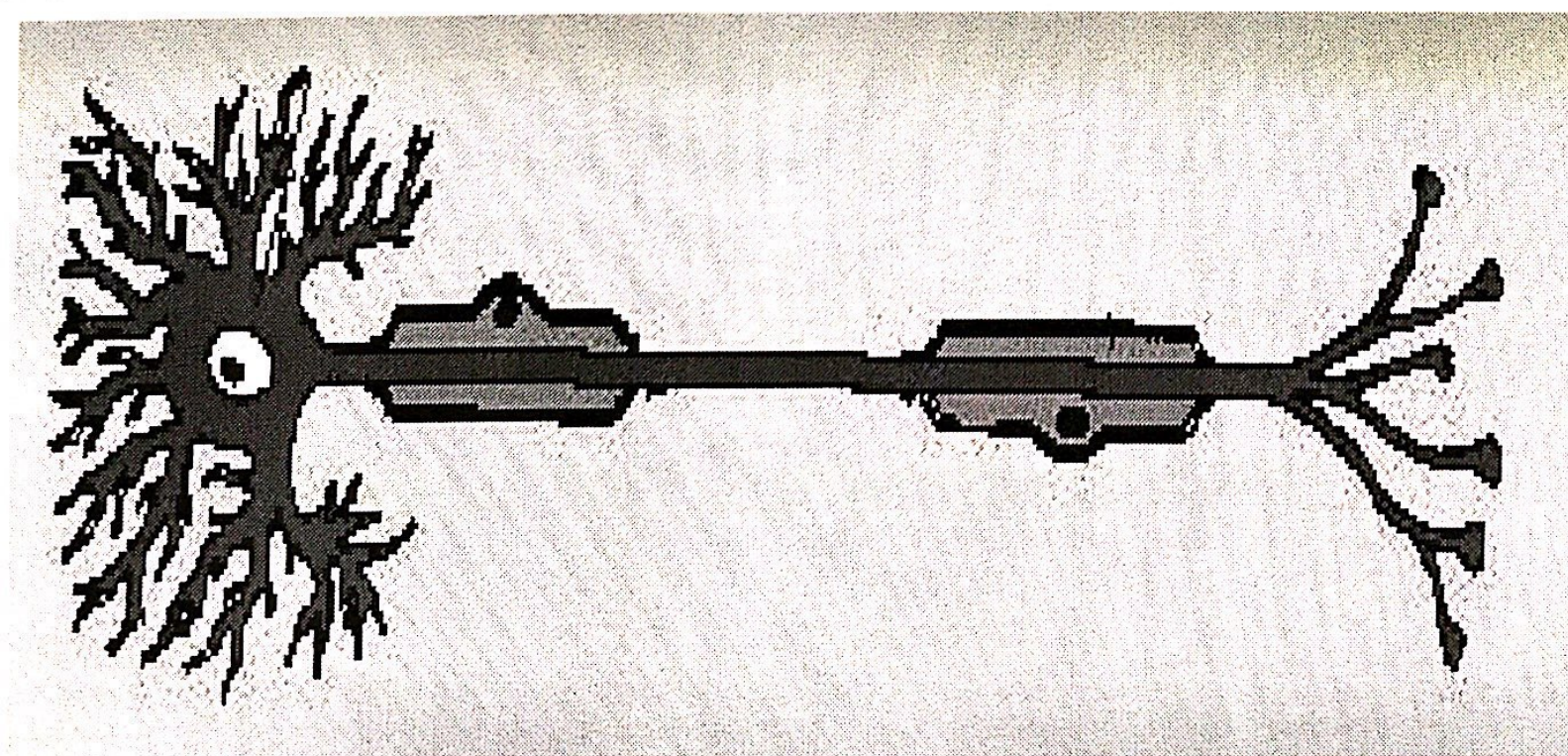
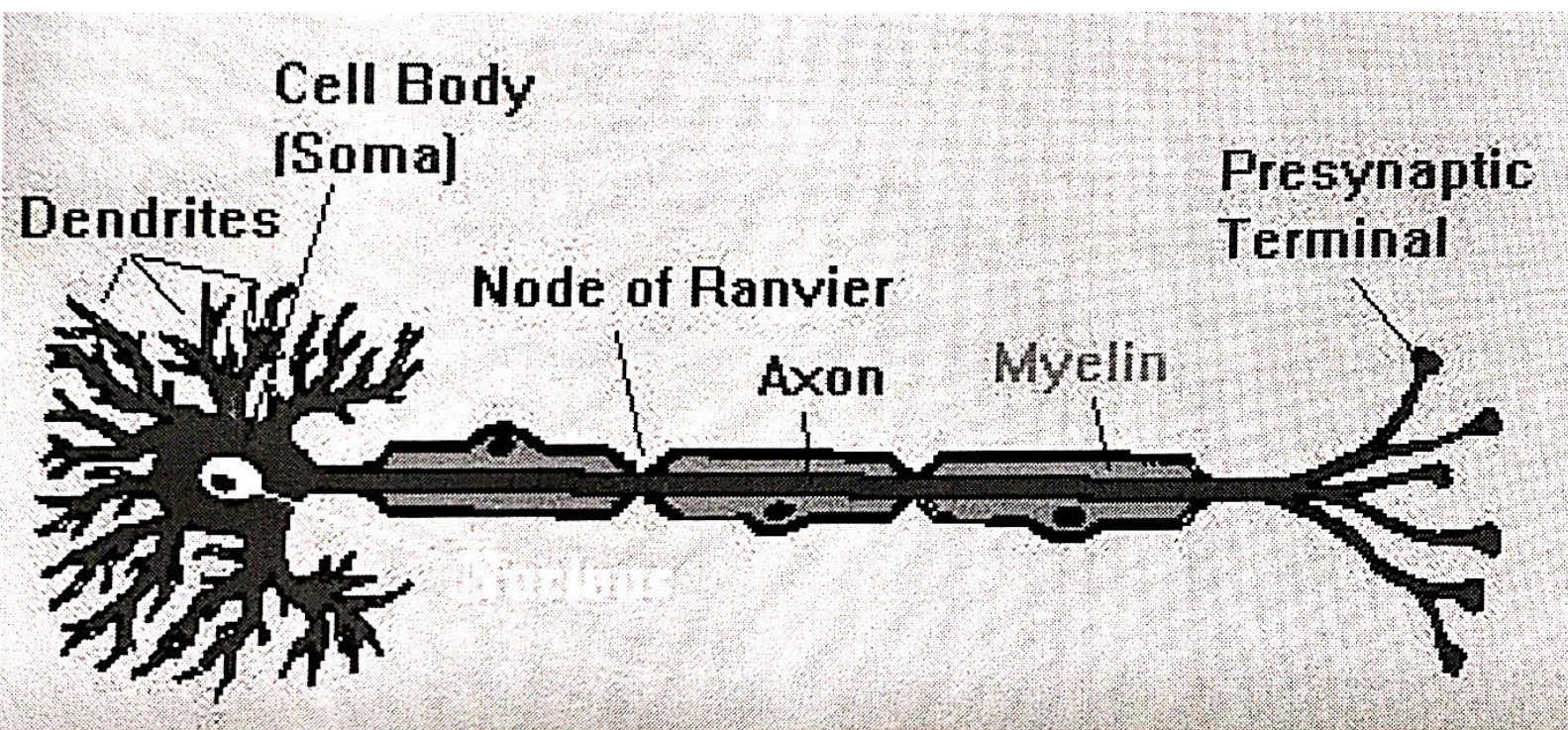
can destruct the myelin sheath - loss of



Multiple sclerosis

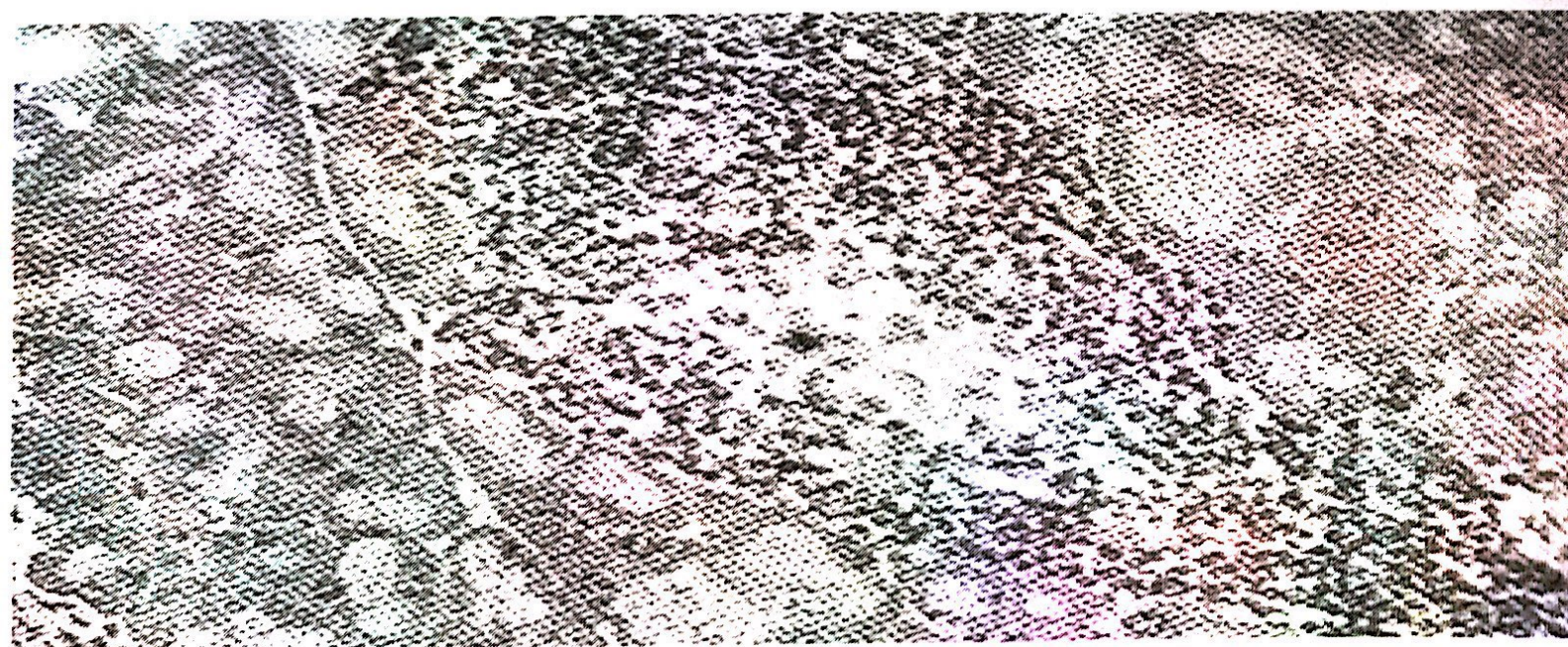
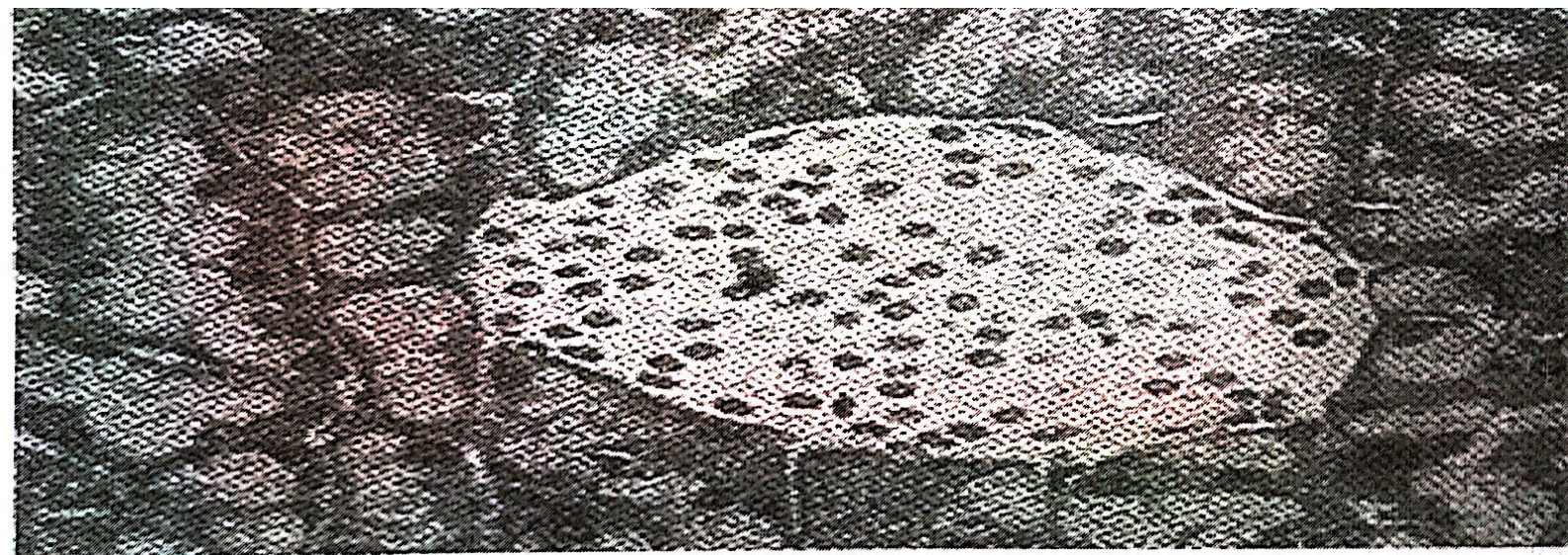
(associated with nervous system)

- Auto T cells destruct the myelin sheath-Loss of coordination
* causes halosination
- Animal model: EAE
- Injection with MBP with CFA cause MS (EAE)
myeline basic protein
- Oral tolerance: ingestion of MBP prior to injection relieve the symptoms in animals It does not work in humans
مصاب
* جانب خارجي
* جانب داخل
* جانب خارجي
* جانب داخل
- Th17 the proinflammatory cells are involved.
- Association with DR2



^{diabetes} Type I IDDM ^(insulin dependant)

- Destruction of β cells of Pancreas (Ag: glu^{glutamic acid decarboxylase} decarboxylase or GAD) and insulin by T cells. so Characterized by insufficient insulin production
- Activation of CD4⁺ (secretion of IL-1, 6, TNF-alpha) and CD8⁺
- Linked to coxsackievirus
- Animal model: NOD mice
- Treated with insulin injections.



Rheumatoid arthritis

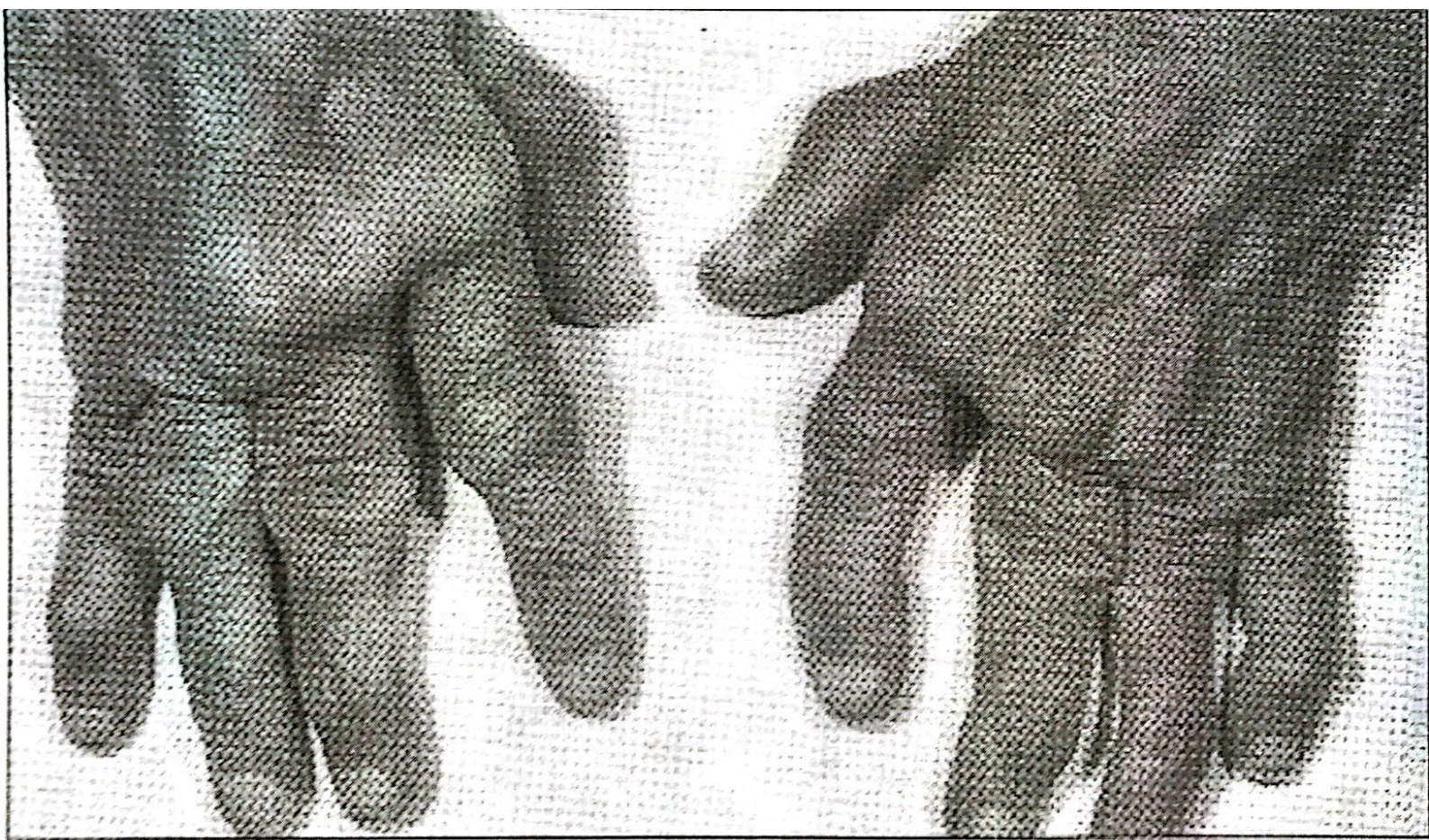
سوء المناعة

- Chronic inflammation of the joint (inflammation of the synovium, dense T, B, NK, mac)-destruction of cartilage and bone-replaced by fibrous tissue.
- RF: Abs (IgM and IgA) against self IgG: minor role
CD4+ Th1: has dominant role
CD4+ Th17 with the inflammatory cytokines: bone and cartilage destruction
- Associated w DR4
- Linked to EBV, rubella and flu
- Mouse → injection w collagen II → RA model
- No cure. Symptoms treated with anti-inflammatory (aspirin) and immunosuppressive drugs.

* synovial fluid in the joints (سوائل المفاصل) there's ~~high~~ IgG and B cells, due to unknown reasons the IgG becomes an antigen that's foreign to the body so the B cells forms IgG, IgM, IgA against that IgG which causes an inflammation of the cartilage → Rheumatoid factors

* For treatment they give monoclonal Ab against TNF (tumor necrosis factor)

* RF factors for detection



Therapeutic strategies

- 1. cytotoxic drugs like cyclophosphamide or azathioprine that interfere w DNA replication
- 2. Blockage of intracellular signals like Cyclosporin A and FK506
- 3. Anticytokine therapy Anti-TNF- α and Anti-IL-1 β Mabs